ABC LOGISTICS PVT. LTD.

Predicting Delivery Delays in Last-Mile Logistics

Case Study for Classroom Distribution — Business Analytics / Operations Management

It was 9:40 p.m. on a Tuesday in July 2025, and the monsoon was hammering the windows of ABC Logistics' dispatch centre in Pune. **Ananya Rao, the company's newly appointed Vice-President of Service Excellence, was still at her desk, staring at a single number on the escalation dashboard: 41.**

Forty-one delivery-delay complaints in a single day from ShopNow alone — the quick-commerce retailer that now accounted for nearly a third of ABC's parcel volume. That morning, ShopNow's category head had been blunt on the call: either ABC's on-time performance improved within the quarter, or the contract would be re-tendered. Ananya had ninety days, a board that wanted a number, and a finance director who flinched at the word "capital expenditure."

The founder, Vikram Mehta, had hired her precisely for moments like this. "We are drowning in data and starving for decisions," he had told her at the interview. Now he wanted an answer to a deceptively simple question: before a parcel even leaves the warehouse, can we tell whether it will be late — and, if so, what should we actually do about it?

ABC Logistics

Founded in 2016, ABC Logistics Pvt. Ltd. had grown from a two-city courier startup into a mid-sized last-mile carrier operating across seven Indian metros. Its promise to clients was simple and absolute: a guaranteed same-day or next-slot delivery window. By 2025, the company was moving roughly 90,000 parcels a day through a network of neighbourhood warehouses, a mixed fleet of vans and two-wheelers, and about 3,500 delivery partners.

The business model lived and died by the delivery window. Clients paid a premium for reliability, and ABC's contracts carried service-level penalties for missed windows. A delayed parcel was therefore expensive twice over: once in the penalty, and again in the slow erosion of the client relationships that penalties signalled. Customer-satisfaction scores tracked on-time performance almost one-for-one.

A Problem Everyone Had an Opinion About

The delay problem was not new, but it had become acute. When Ananya pulled a representative sample of recent deliveries, she found that roughly half had missed their promised window — a figure that stopped the conversation during the Monday operations review. What the room could not agree on was the cause.

Rohit Deshpande, the VP of Fleet & Operations and a fifteen-year veteran of the industry, had a firm view.

> *"It's the drivers and it's the roads. Give me budget to hire experienced riders and to add vehicles, and the delays go away. The new hires don't know the routes, they take too many stops, and half our streets are dug up. That's the whole story."*
>
> — Rohit Deshpande, VP Fleet & Operations

Meera Iyer, the finance director, was unconvinced that the answer was more spending.

> *"Every problem in this company somehow ends in a request for capital. Before I sign off on new vans or a hiring spree, I want to see evidence that those are the things driving delays — not just the things that are easiest to blame."*
>
> — Meera Iyer, Finance Director

Ananya had her own suspicion, one she kept to herself for the moment. In her first weeks, she had spent time on the warehouse floor and had been struck by how long parcels sometimes sat between arrival and dispatch — waiting for manifests, for sorting, for a vehicle to fill. Nobody measured that wait as a cause of delay; it was treated as "just how the warehouse works." But she had no proof, and proof was what the board would demand.

The Analytics Initiative

Rather than arguing from anecdote, Ananya asked her analyst, Kavya Nair, to assemble a clean dataset to settle the question empirically. Over two weeks, Kavya compiled records for 1,000 completed deliveries, spanning warehouses, time slots, weather conditions, and vehicles. For each delivery, she recorded eleven operational and environmental factors captured at or before dispatch, together with the outcome that mattered: whether the delivery ultimately missed its window.

The eleven factors spanned everything the operations review had argued about — distance and traffic, weather and road quality, the driver's experience, the number of stops on the route, the age of the vehicle and its fuel efficiency, the weight of the package, and, at Ananya's insistence, the time the parcel spent being processed inside the warehouse before it was dispatched. The full data dictionary appears in Exhibit 1, a sample of the records in Exhibit 2, and summary statistics in Exhibit 3.

Crucially, every factor was something ABC could know before the van pulled out of the yard. If a reliable pattern existed, the company could score each parcel at dispatch, flag the ones most at risk, and intervene — reassign a driver, re-sequence a route, or warn the customer — while there was still time to act.

The Decision

Ananya's brief to the analytics team was therefore two-sided, and she was careful to separate the two halves. The first was a prediction problem: build a model that, using only information available at dispatch, estimates the probability that a given delivery will be delayed, and does so accurately enough to be trusted on parcels it has never seen.

The second was the harder half — the management problem. A probability is not a plan. The board would not fund "a model"; it would fund actions. Ananya needed the analysis to tell her which factors genuinely drove delays and, among those, which ones ABC could actually influence — and at what cost relative to the benefit. Rohit's instinct pointed at drivers and vehicles; Meera's scepticism pointed at proving the case before spending; Ananya's own hunch pointed at the warehouse. Within ninety days, one of them — or the data — would be shown right.

She had the dataset, an analyst, and a deadline. What she did not yet have was an answer.

Questions

Q1. Build a model to predict whether a delivery will be delayed.

Q2. Evaluate the model?

Q3. Make an application for delay prediction for the supervisor or manager ?

EXHIBITS

Exhibit 1. Data Dictionary

Variable	Description	Type / Scale
Delivery_Distance	Distance from warehouse to customer	Continuous (km)
Traffic_Congestion	Congestion on the route, 1 (low) to 5 (high)	Ordinal 1–5
Weather_Condition	1 = Clear, 2 = Rainy, 3 = Stormy	Categorical
Delivery_Slot	1 = Morning, 2 = Afternoon, 3 = Evening	Categorical
Driver_Experience	Driver's delivery experience	Continuous (years)
Num_Stops	Stops before the final delivery	Count
Vehicle_Age	Age of the delivery vehicle	Continuous (years)
Road_Condition_Score	Road quality, 1 (poor) to 5 (excellent)	Ordinal 1–5
Package_Weight	Weight of the package	Continuous (kg)
Fuel_Efficiency	Vehicle fuel efficiency	Continuous (km/l)
Warehouse_Processing_Time	Time to process & dispatch at warehouse	Continuous (min)
Delivery_Delay	Outcome: 1 = delayed, 0 = on time	Binary (target)

Exhibit 2. Sample of Delivery Records (first 12 of 1,000)

Dist	Traf	Wthr	Slot	DrvExp	Stops	VehAge	RoadSc	PkgWt	FuelEff	WhProc	DELAY
19.4	4	3	2	16	6	9	3	33.6	13.0	58	1
47.6	3	2	1	10	4	2	3	20.8	15.1	26	0
36.9	5	3	2	18	3	9	3	23.8	11.3	72	1
30.3	1	2	3	6	7	8	2	2.9	16.0	64	1
8.6	5	1	1	19	2	9	2	10.0	12.8	32	1
8.6	1	3	3	4	6	4	5	39.6	6.1	43	0
3.8	2	1	1	13	4	1	2	23.0	7.9	114	1
43.4	1	3	1	14	6	1	2	15.4	17.6	114	1
30.5	4	1	1	3	4	5	2	19.5	15.9	78	0
35.7	3	3	1	10	4	2	4	42.8	18.4	24	0
2.0	2	1	2	6	4	8	2	37.6	6.8	56	1
48.5	1	1	3	1	9	5	1	14.0	19.9	12	0

Exhibit 3. Summary Statistics (n = 1,000)

Variable	Mean	Std. Dev.	Min	Max
Delivery_Distance (km)	25.0	14.3	1.2	50.0
Traffic_Congestion (1–5)	3.0	1.4	1	5

Variable	Mean	Std. Dev.	Min	Max
Weather_Condition (1–3)	2.0	0.8	1	3
Delivery_Slot (1–3)	2.0	0.8	1	3
Driver_Experience (yrs)	10.4	5.3	1	19
Num_Stops	5.0	2.6	1	9
Vehicle_Age (yrs)	4.9	2.5	1	9
Road_Condition_Score (1–5)	3.0	1.4	1	5
Package_Weight (kg)	24.7	14.1	0.5	50.0
Fuel_Efficiency (km/l)	12.3	4.3	5.0	20.0
Warehouse_Processing_Time (min)	64.4	32.2	10	119

— End of Case —

This case was prepared as the basis for classroom discussion. ABC Logistics, its personnel, and the events described are fictional; the dataset is provided for instructional use.